PROPERNESS OF THE KEEL-MORI MODULI MAP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4283-4294 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4296-4305, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 49 of queue_stacks-stacks-more-morphisms.json (pb=388, stmt 4283-4294, proof 4296-4305).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-keel-mori-diagonal: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4245–4254. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-keel-mori-finite-type: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4204–4211. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma theorem-keel-mori: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3854–3862. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $p : \mathcal{X} \to Y$ be a morphism from an algebraic stack to an algebraic space. Assume*

(1) *$\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is finite,*
(2) *p is proper, and*

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(3) *Y is locally Noetherian.*

Let $f : \mathcal{X} \to M$ be the moduli space constructed in Theorem theorem-keel-mori. Then $M \to Y$ is proper.

Proof. By Lemma lemma-keel-mori-finite-type we see that $M \to Y$ is locally of finite type. By Lemma lemma-keel-mori-diagonal we see that $M \to Y$ is separated. Of course $M \to Y$ is quasi-compact and universally closed as these are topological properties and $\mathcal{X} \to Y$ has these properties and $\mathcal{X} \to M$ is a universal homeomorphism. □